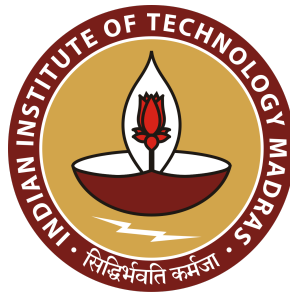


GRAND CHALLENGE - 2025

TOOLS INSTALLATION IN UBUNTU



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0.1 Proprietary Notice

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0.2 Release Information

Version	Date	Updates	Reviewed by
1.0	16/09/2025	Vignesh K	Kotteeswaran E
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1. Introduction:

This document details the hardware and software tools being needed for Shakti SoC development. The hardware setup with Ubuntu 22.04 needs the following .

- Vivado IDE
- Vivado Cable driver.
- Copying board files.
- Bluespec Compiler.
- GTKWAVE

The software dependencies are

- RISC-V GNU toolchain.
- Spike
- Verilator.
- Gtkterm

2. Common Dependencies:

Install these dependencies which are common in some of them.

```
sudo apt-get install autoconf automake autotools-dev curl make libmpc-dev  
libmpfr-dev libgmp-dev bc gawk build-essential bison flex texinfo gperf libtool  
patchutils zlib1g-dev pkg-config
```

3. Hardware dependencies

Install the following dependencies before proceeding with any other installation. Copy the table 1 and paste the same in a terminal.

3.1. Install dependencies

Install the following dependencies by running it in the ubuntu terminal.

```
sudo apt-get install libusb-1.0-0-dev libusb-0.1-4 libftdi1 libftdi1-2 libncurses5
```

3.2. Vivado Setup:

- a. Download Vivado SDK edition from the AMD website.

```
https://www.xilinx.com/member/forms/download/xef-vivado.html?filename=Xilinx\_Vivado\_SDK\_2018.3\_1207\_2324.tar.gz
```

A login is required to download the tar file in the AMD website.

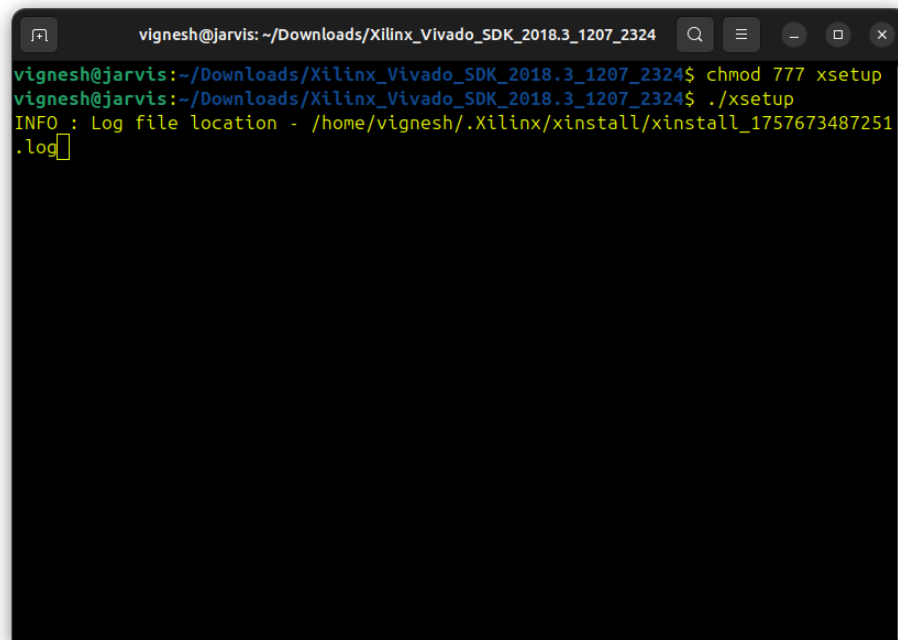
- b. Untar the file "<filename>.tar.xz"

```
tar xf Xilinx_Vivado_SDK_2018.3_1207_2324.tar.gz
```

- c. Give write permission to the file "xsetup" by using the command

```
chmod 777 xsetup
```

d. Run the code “./xsetup” in the terminal”



```
vignesh@jarvis: ~/Downloads/Xilinx_Vivado_SDK_2018.3_1207_2324
vignesh@jarvis:~/Downloads/Xilinx_Vivado_SDK_2018.3_1207_2324$ chmod 777 xsetup
vignesh@jarvis:~/Downloads/Xilinx_Vivado_SDK_2018.3_1207_2324$ ./xsetup
INFO : Log file location - /home/vignesh/.Xilinx/xinstall/xinstall_1757673487251.log
```

Figure 1: Installation Setup command

Installation flow:

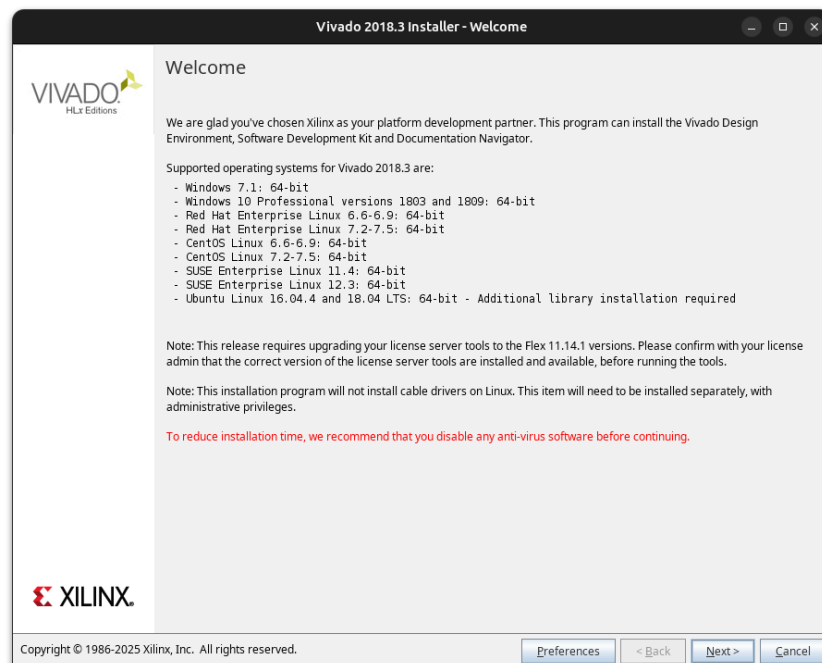


Figure 2: Give next and proceed.

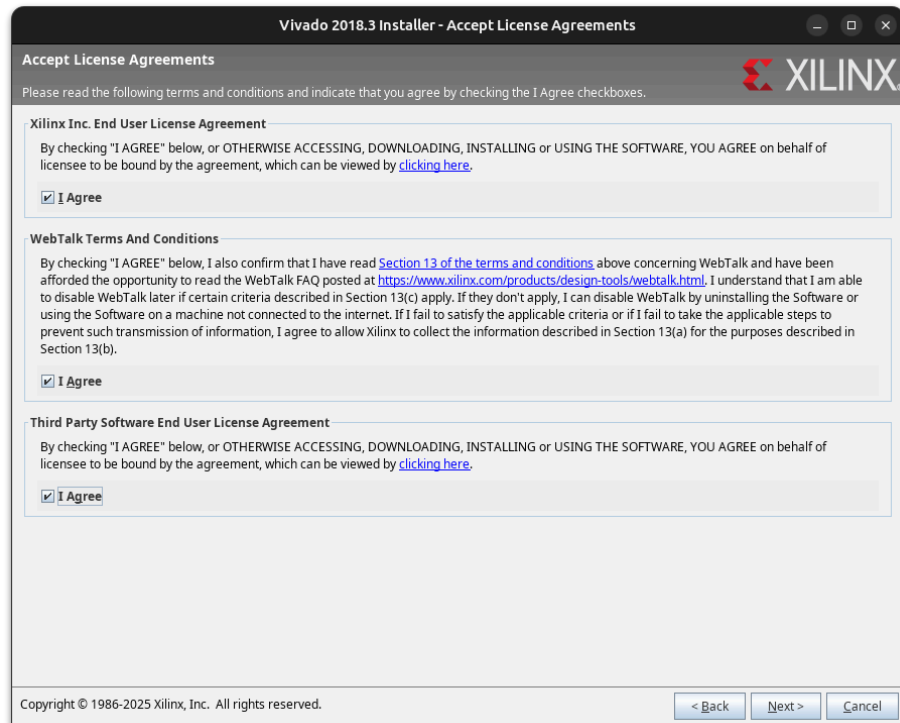


Figure 3: Agree to everything and proceed.

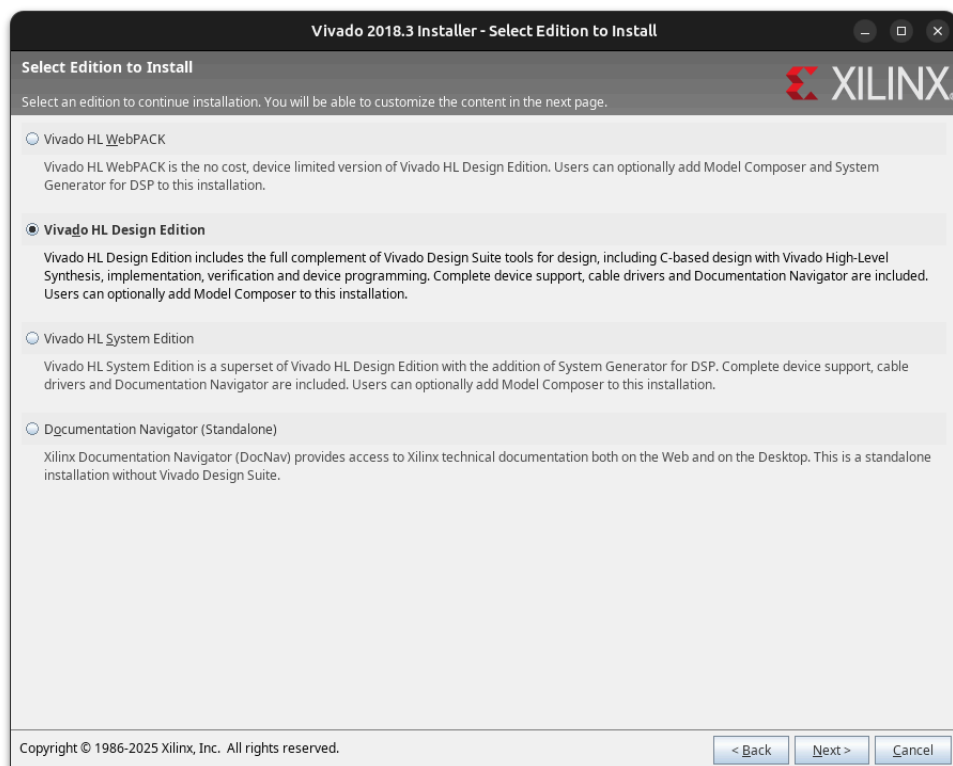


Figure 4: Select Vivado HL Design edition and proceed.

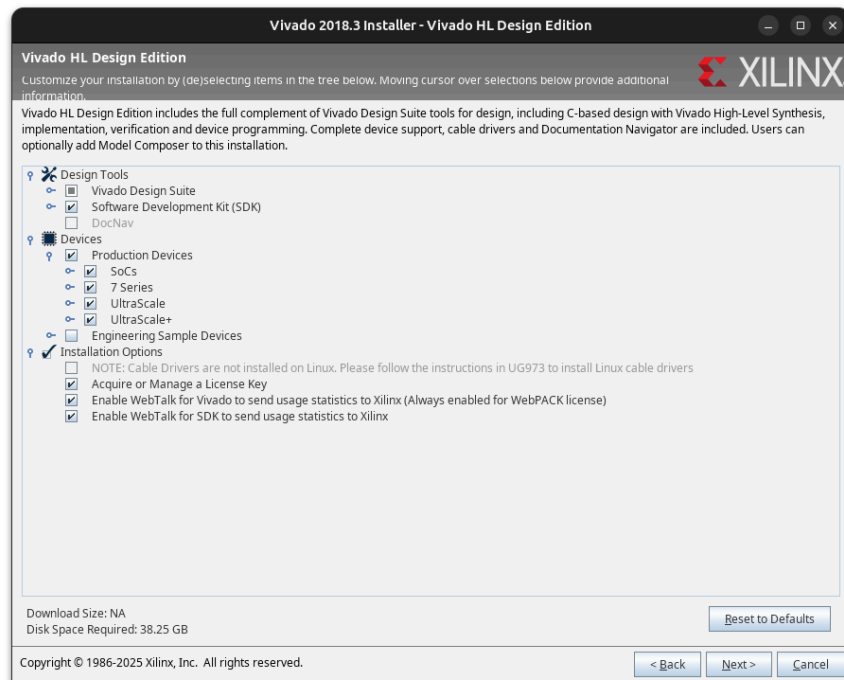


Figure 5: Select Everything and proceed.

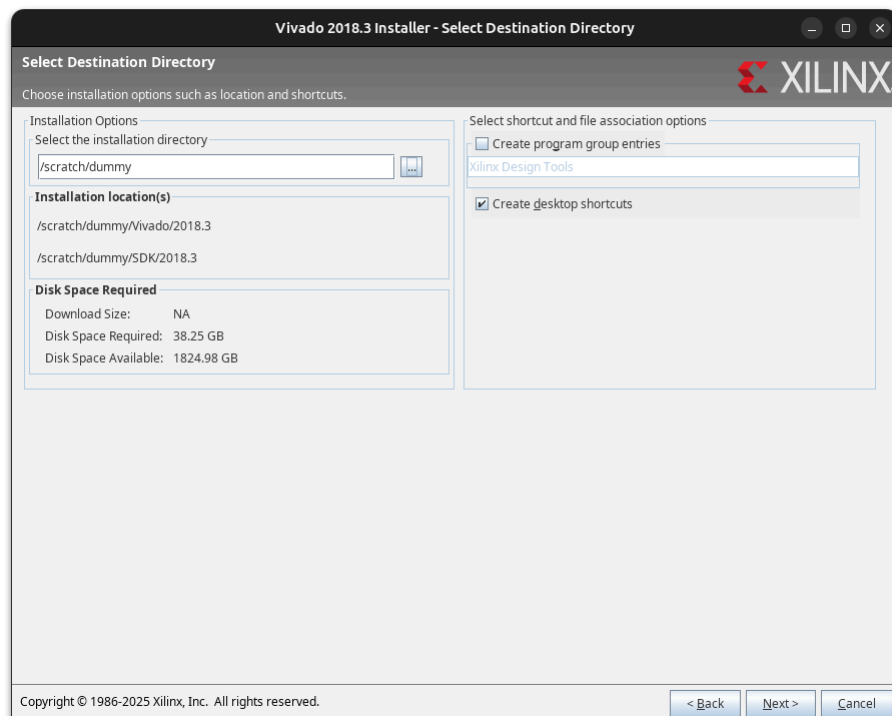


Figure 6: Enter the directory wherever Vivado is to be installed and deselect this create program group entries box and then proceed.

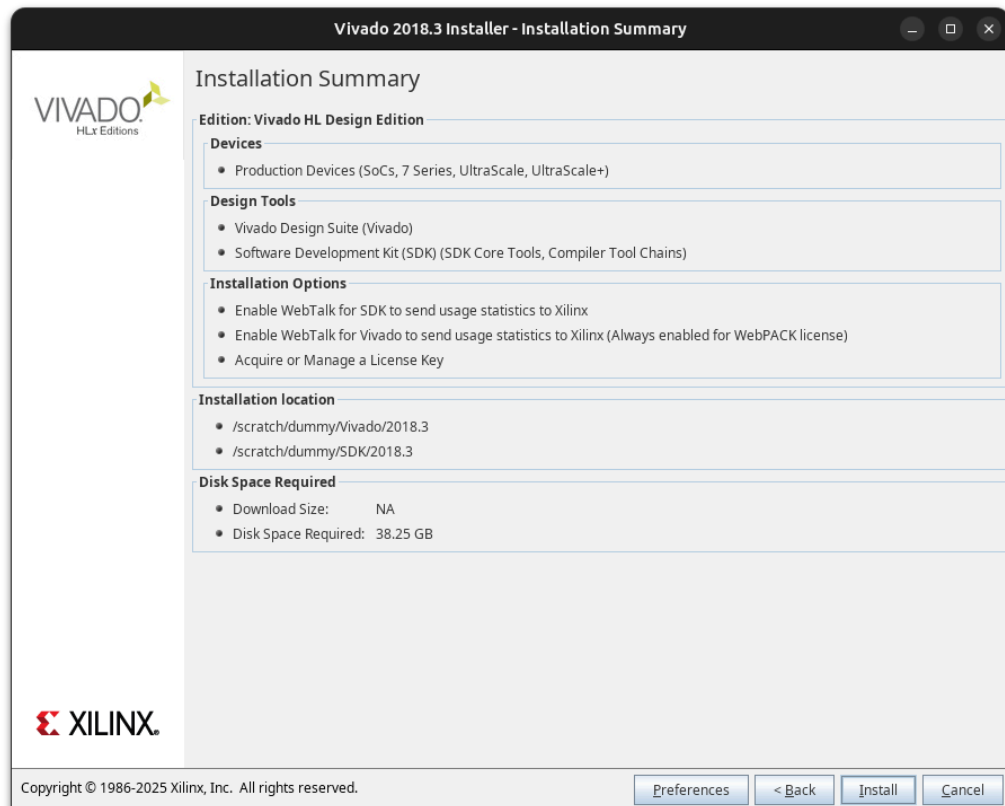


Figure 7: At this stage, install and proceed.

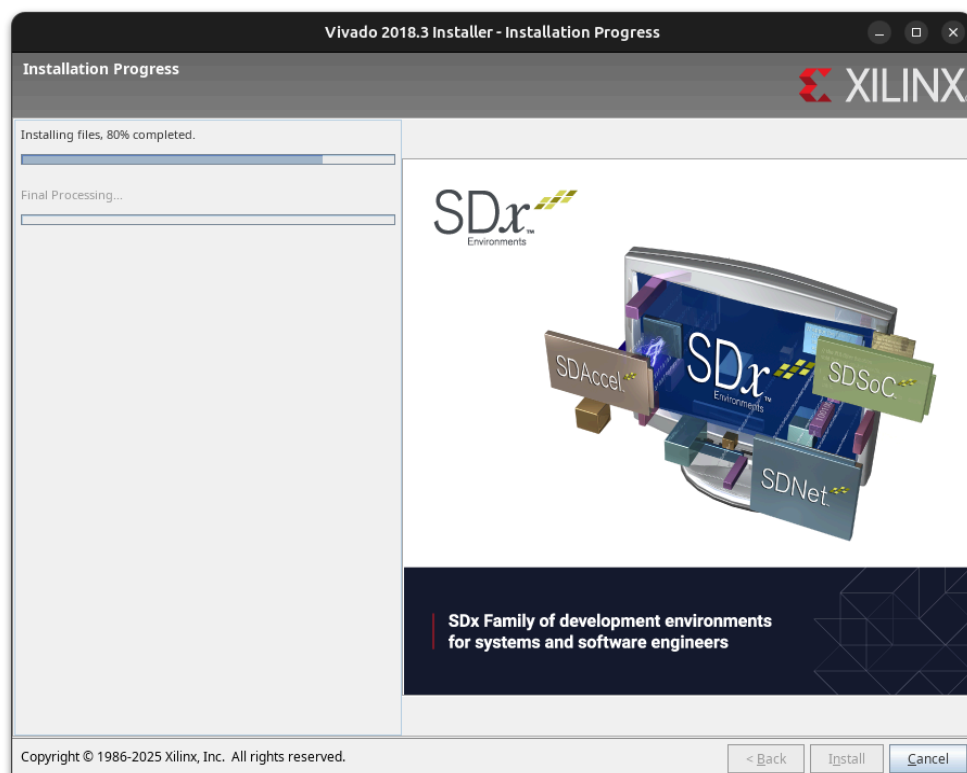
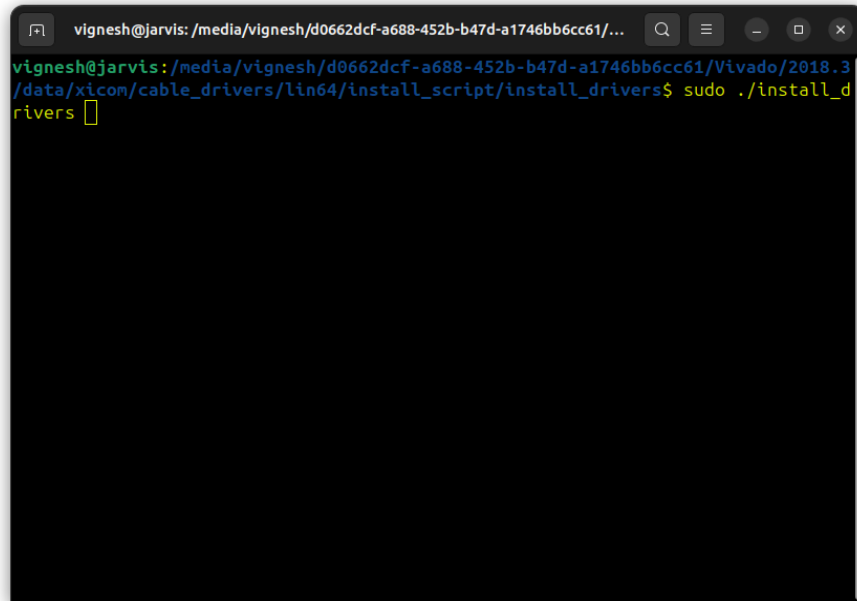


Figure 8: Vivado Installation Window

e. After installation, do the following.

- Go to the folder "[installed folder]/vivado/Vivado/2018.3/data/xicom/cable_drivers/lin64/install_script/install_drivers/"
- run "sudo ./install_drivers"

A terminal window with a dark background. The title bar shows the path "/media/vignesh/d0662dcf-a688-452b-b47d-a1746bb6cc61/...". The prompt is "vignesh@jarvis:". The command entered is "sudo ./install_drivers" and the cursor is at the end of the line.

```
vignesh@jarvis: /media/vignesh/d0662dcf-a688-452b-b47d-a1746bb6cc61/...
vignesh@jarvis:/media/vignesh/d0662dcf-a688-452b-b47d-a1746bb6cc61/Vivado/2018.3/data/xicom/cable_drivers/lin64/install_script/install_drivers$ sudo ./install_drivers
```

Figure 9: Drivers Installation command

These drivers are not available pre-installed in the Vivado ,so it is required to install them before connecting the board.

3.3. Adding board files

- a. Board files can be added on cloning the board files from digilent's github page.
- b. In this GitHub repository, there are board files for almost every Digilent Boards.

```
git clone https://github.com/Digilent/vivado-boards/tree/master/new/board_files
```

- c. Add the cloned board files from Github which is required to /path/to/Vivado/2018.3/data/boards/board_files.

3.4. Invoking Vivado application

Vivado application can be invoked in two following ways.

1.Through icon in home:

- 1. Go to the Desktop and find the vivado icon
- 2. Right click and select "Allow Launch"
- 3.Double clicking the icon should open Vivado

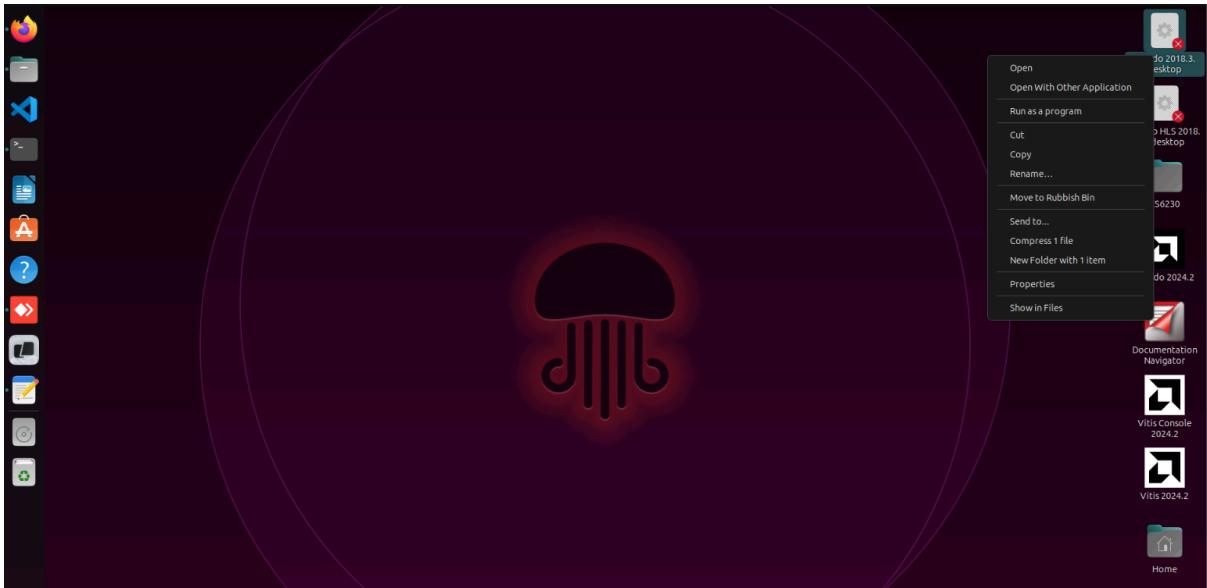


Figure 10: Opening Vivado through icon

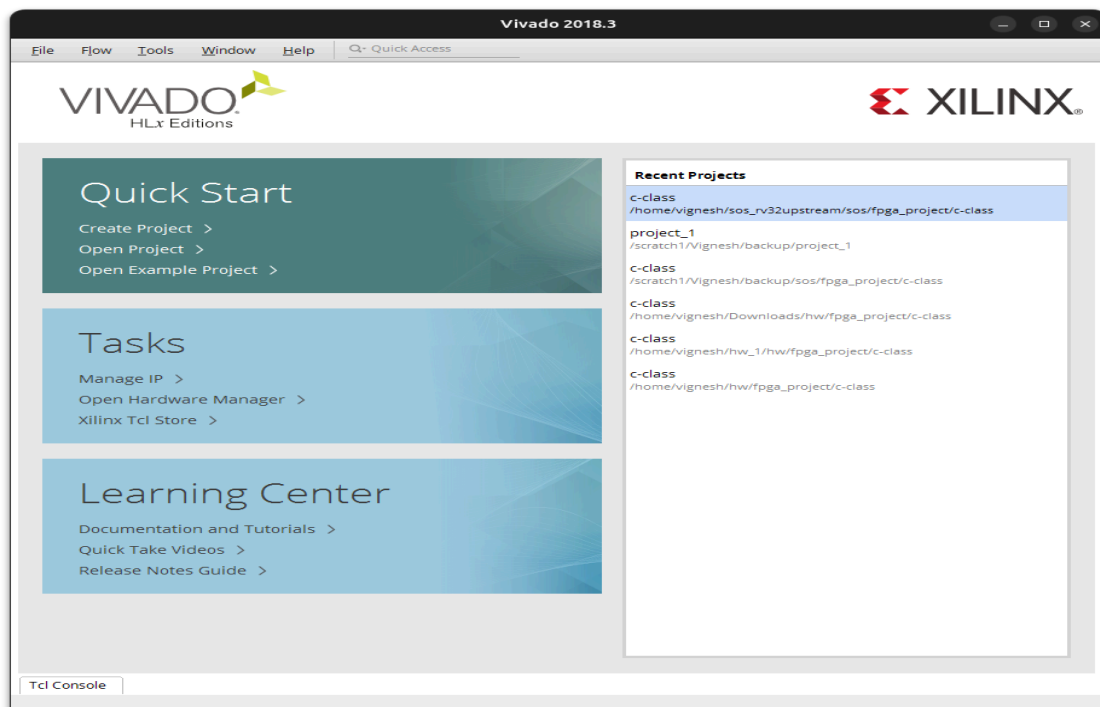
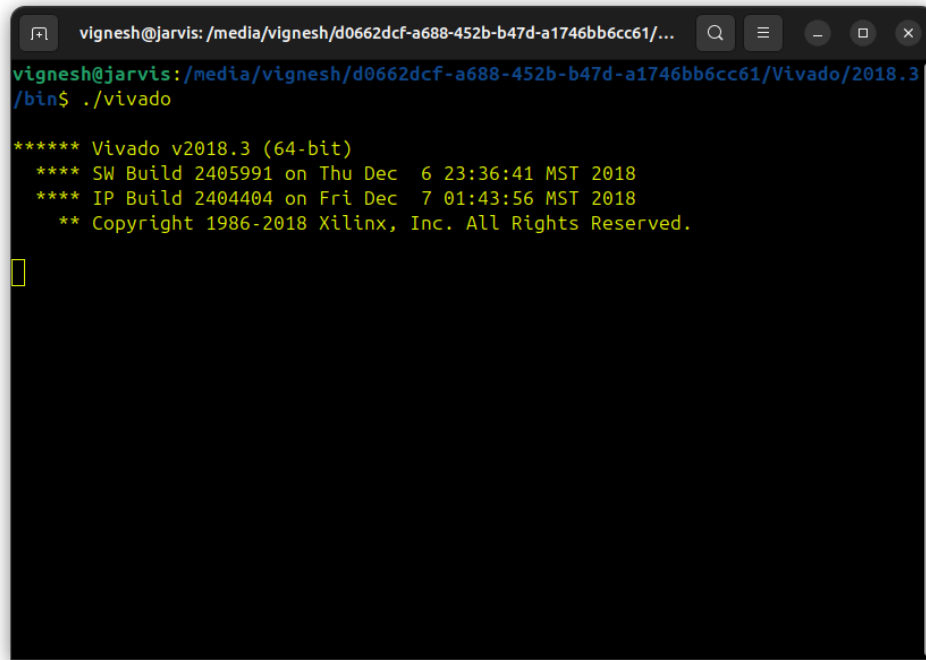


Figure 11: Opening Vivado

2.Through Terminal

- Go to following path “/path/install/vivado/Vivado/2018.3/”
Right click and open new terminal
- Run the following command `./vivado` , this should open vivado



```
vignesh@jarvis: /media/vignesh/d0662dcf-a688-452b-b47d-a1746bb6cc61/...  
vignesh@jarvis:/media/vignesh/d0662dcf-a688-452b-b47d-a1746bb6cc61/Vivado/2018.3  
/bin$ ./vivado  
  
***** Vivado v2018.3 (64-bit)  
**** SW Build 2405991 on Thu Dec 6 23:36:41 MST 2018  
**** IP Build 2404404 on Fri Dec 7 01:43:56 MST 2018  
** Copyright 1986-2018 Xilinx, Inc. All Rights Reserved.  
  
█
```

Figure 12: Opening Vivado through terminal

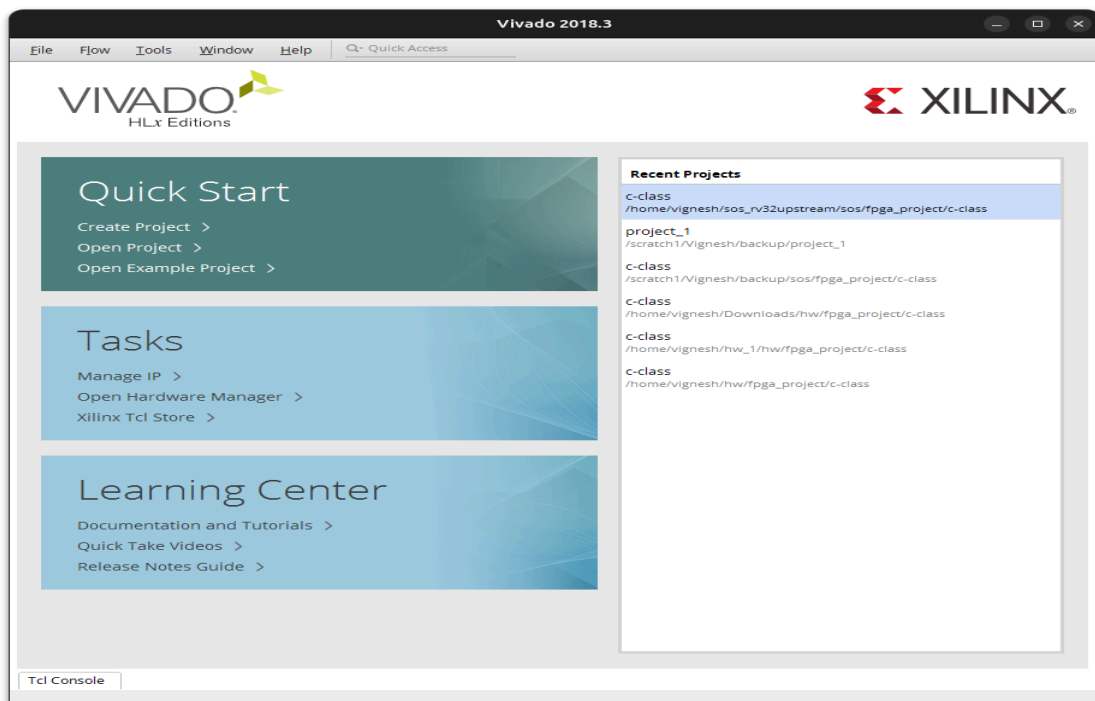


Figure 13: Opening Vivado

4. Installing Bluespec complier:

4.1. Install the dependencies:

```
sudo apt-get install ghc libghc-regex-compat-dev libghc-syb-dev  
libghc-old-time-dev libghc-split-dev tcl-dev
```

4.2. Clone this repository:

```
git clone --recursive https://github.com/B-Lang-org/bsc
```

4.3. Change directory to bsc and checkout to 2025.01.1 tag.

```
cd bsc  
git checkout 2025.01.1
```

4.4. Install Bluespec compiler from source:

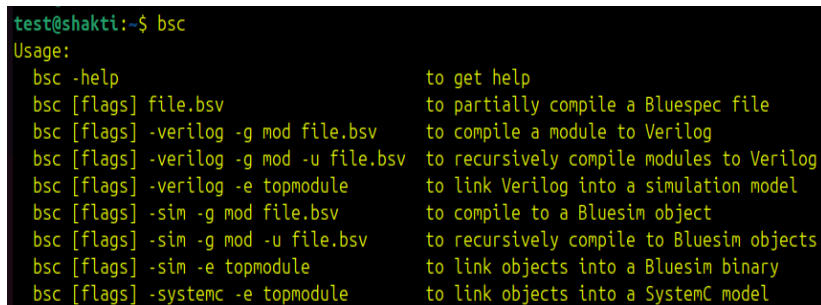
```
sudo make install-src
```

4.5. Add BSC to PATH with this command:

```
echo 'export PATH=/home/test/bsc/inst/bin:$PATH' >> ~/.bashrc
```

4.6. It can be invoked by these commands:

```
bsc
```



```
test@shakti:~$ bsc  
Usage:  
bsc -help                to get help  
bsc [flags] file.bsv      to partially compile a Bluespec file  
bsc [flags] -verilog -g mod file.bsv  to compile a module to Verilog  
bsc [flags] -verilog -g mod -u file.bsv to recursively compile modules to Verilog  
bsc [flags] -verilog -e topmodule      to link Verilog into a simulation model  
bsc [flags] -sim -g mod file.bsv       to compile to a Bluesim object  
bsc [flags] -sim -g mod -u file.bsv     to recursively compile to Bluesim objects  
bsc [flags] -sim -e topmodule          to link objects into a Bluesim binary  
bsc [flags] -systemc -e topmodule      to link objects into a SystemC model
```

Figure 14: Invoking Bluespec Compiler

5. Installing GTKWAVE:

5.1. Install gtkterm using the command:

```
sudo apt install gtkwave -y
```

6. Software Dependencies:

6.1. Install dependencies.

```
sudo apt-get install python3 python3-pip libexpat-dev ninja-build git cmake  
libglib2.0-dev libslirp-dev
```

6.2. Install RISC-V-GNU-TOOLCHAIN:

```
git clone https://github.com/riscv/riscv-gnu-toolchain
```

6.3. After cloning go into the folder , update the submodule, change version of gcc to 13.2 ,configure and make.

```
cd riscv-gnu-toolchain  
git submodule update --init --recursive  
sudo rm -rf gcc  
git clone https://gcc.gnu.org/git/gcc.git  
cd gcc  
git checkout releases/gcc-13.2.0  
cd ..  
mkdir build_rv  
./configure --prefix=[Path to risc-v toolchain]/riscv-gnu-toolchain/build_rv  
--enable-multilib  
sudo make -j $(nproc)
```

These commands will configure ,make and install the toolchain. This will contain both 32-bit and 64-bit toolchain

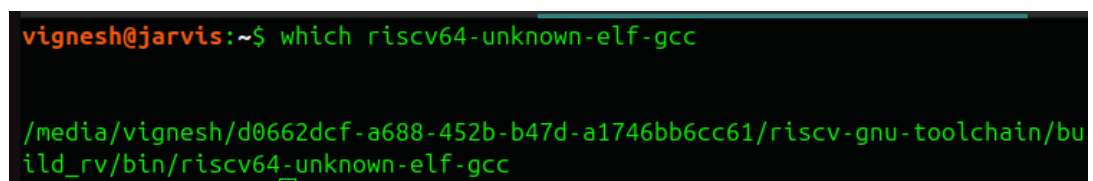
6.4. Add path to .bashrc with the following commands:

(This will by default invoke the toolchain)

```
echo 'export PATH=[Path to risc-v  
toolchain]/riscv-gnu-toolchain/build_rv/bin:$PATH' >> ~/.bashrc
```

6.5. To verify try these commands:

```
which riscv64-unknown-elf-gcc  
  
riscv64-unknown-elf-gcc --print-multi-lib
```



```
vignesh@jarvis:~$ which riscv64-unknown-elf-gcc  
  
/media/vignesh/d0662dcf-a688-452b-b47d-a1746bb6cc61/riscv-gnu-toolchain/bu  
ild_rv/bin/riscv64-unknown-elf-gcc
```

Figure 15: Checking the path of multilib RISC-V toolchain

```
vignesh@jarvis:~$ riscv64-unknown-elf-gcc --print-multi-lib
.;
rv32i/ilp32;@march=rv32i@mabi=ilp32
rv32im/ilp32;@march=rv32im@mabi=ilp32
rv32iac/ilp32;@march=rv32iac@mabi=ilp32
rv32imac/ilp32;@march=rv32imac@mabi=ilp32
rv32imafc/ilp32f;@march=rv32imafc@mabi=ilp32f
rv64imac/lp64;@march=rv64imac@mabi=lp64
rv64imafdc/lp64d;@march=rv64imafdc@mabi=lp64d
```

Figure 16: Checking the invocation of all other toolchain

7. Installing SPIKE:

7.1. Install Dependencies:

```
sudo apt-get install device-tree-compiler libboost-regex-dev libboost-system-dev
```

7.2. Clone the RISC_V_ISA_SIM repository:

```
git clone https://github.com/riscv-software-src/riscv-isa-sim.git
```

7.3. Go into the cloned directory and checkout to the given git.

```
cd riscv-isa-sim
git checkout d39cbb3ad27b2de882260c0a22f11a59efebbb151
```

This will checkout the repository to the specified branch.

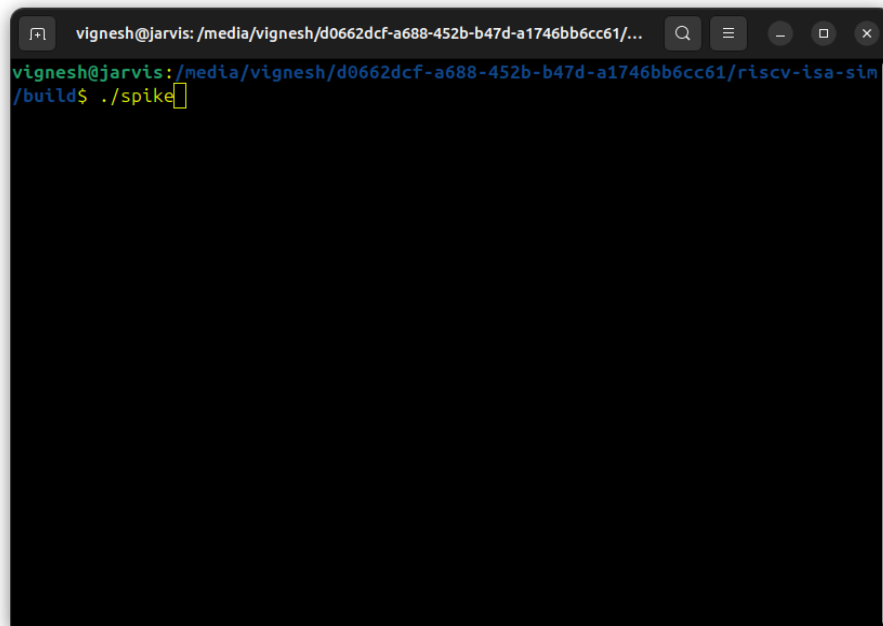
7.4. Create a directory named build, go inside, configure and make.

```
mkdir build
cd build
export RISC_V=/path/to/install/spike
./configure --prefix=$RISC_V
sudo make -j $(nproc)
sudo make install
```

These commands will configure ,make and install spike.

7.5. Spike can be invoked by giving these commands:

```
./spike
```



```
vignesh@jarvis: /media/vignesh/d0662dcf-a688-452b-b47d-a1746bb6cc61/...
vignesh@jarvis:/media/vignesh/d0662dcf-a688-452b-b47d-a1746bb6cc61/riscv-isa-sim
/build$ ./spike
```

Figure 17: Invoking Spike

8. Installing VERILATOR:

8.1. Install the dependencies:

```
sudo apt-get install perl g++ ccache libgoogle-perftools-dev numactl perl-doc libfl2
libfl-dev zlib1g help2man
```

8.2. Clone this repository:

```
git clone https://github.com/verilator/verilator
```

8.3. Change directory to bsc and checkout to 2025.01.1 tag.

```
cd verilator
git checkout v5.024
```

This will checkout to the branch v5.024

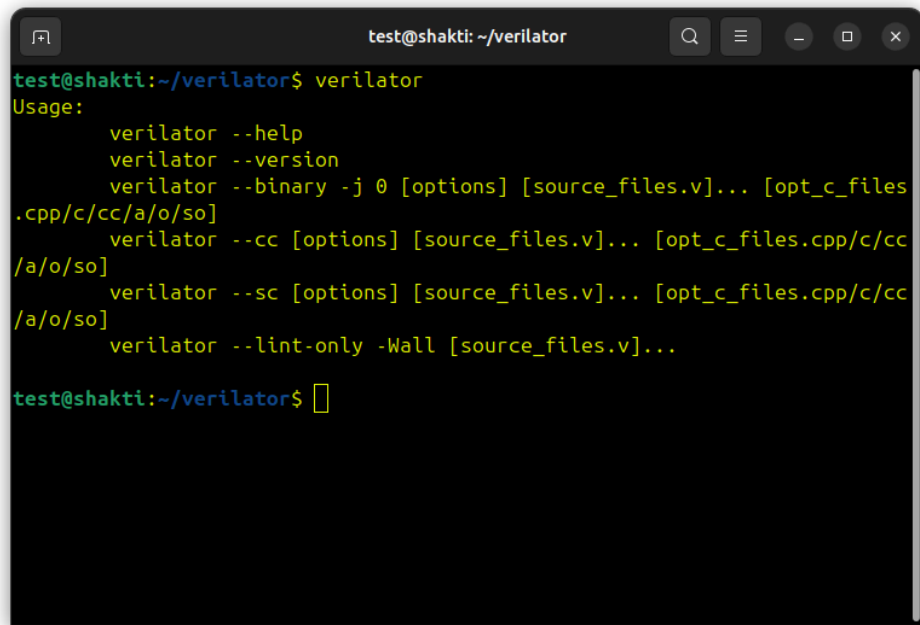
8.4. Install Verilator:

```
unset VERILATOR_ROOT
autoconf
./configure
make -j `nproc`
sudo make install
```

These commands will configure ,make and install verilator.

8.5. It can be invoked by these commands:

```
verilator
```


A terminal window titled 'test@shakti: ~/verilator' showing the output of the 'verilator' command. The output displays the usage of Verilator, including options like --help, --version, --binary, --cc, --sc, and --lint-only. The prompt 'test@shakti:~/verilator\$' is visible at the bottom.

```
test@shakti:~/verilator$ verilator
Usage:
    verilator --help
    verilator --version
    verilator --binary -j 0 [options] [source_files.v]... [opt_c_files
.cpp/c/cc/a/o/so]
    verilator --cc [options] [source_files.v]... [opt_c_files.cpp/c/cc
/a/o/so]
    verilator --sc [options] [source_files.v]... [opt_c_files.cpp/c/cc
/a/o/so]
    verilator --lint-only -Wall [source_files.v]...

test@shakti:~/verilator$
```

Figure 18: Checking Verilator through command

9. Installing GTKTerm

9.1. Install gtkterm using the command:

```
sudo apt install gtkterm -y
```

9.2. Opening gtkterm:

```
sudo gtkterm
```

This completes the tools setup for both hardware build and software build.